Cell Signalling: An Overview

- Cells communicate through chemical messengers that bind specific receptors.
- Signal transduction converts an external cue into an internal response.
- Amplification means one ligand can trigger thousands of downstream events.
- Termination matters as much as activation: signals must switch off.

Four Families of Receptor

- G-protein coupled receptors: seven transmembrane passes, slow and versatile.
- Receptor tyrosine kinases: dimerise on binding, then autophosphorylate.
- Ion channel receptors: fastest response, measured in milliseconds.
- Nuclear receptors: lipophilic ligands act directly on transcription.

Second Messengers

- cAMP is produced by adenylyl cyclase and activates protein kinase A.
- Calcium is stored in the endoplasmic reticulum and released by IP3.
- Diacylglycerol stays in the membrane and recruits protein kinase C.
- Each messenger has a dedicated clearance route, which sets signal duration.

Break

- Five minutes. Then: feedback loops.

Feedback and Robustness

- Negative feedback stabilises output against fluctuating input.
- Positive feedback creates switches: once tripped, the state persists.
- Cross-talk lets two pathways share components, at the cost of specificity.
- Disease often follows from feedback that cannot switch off.

Summary

- Receptor family determines speed; second messenger determines reach.
- Amplification and termination together define the shape of a response.
- Read chapter 15 before the seminar.